

Data-Driven Multiple ARIMA through Neural Fusion for Enhanced Wind Power Prediction with Missing Data

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Background

1. The ARIMA model is a well-recognized linear model often valued for its precision in time series forecasting.
2. While fusing ARIMA and neural networks (NNs) is an effective strategy to capture both linear and nonlinear data features, the combined approach remains sensitive to missing data.
3. Wind energy prediction is a typical application of time series forecasting.

Motivations

1. Missing data severely compromises the reliability of Wind Power Prediction.
2. Data is available from distinct operational regimes within wind energy systems.
3. Employing a multiple model fusion mechanism can significantly enhance forecasting accuracy.

Multiple ARIMA fusion framework

ARIMA Modeling: $$\left(1-\sum_{k=1}^{p} a_i\, z^{-k}\right)(1-z^{-1})^d y_t = \left(1+\sum_{k=1}^{q} b_j\, z^{-k}\right)\varepsilon_t$$

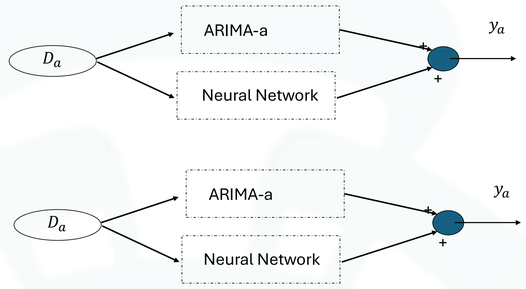
$$\theta = [YY^T]^{-1}Y^T Y$$

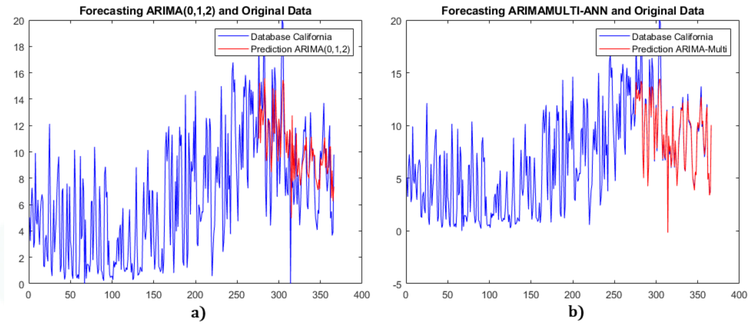
NN Training: $[V^*, W^*] = argmin_{V,W} \sum e_i^2(k+1) + \lambda \sum_i \|W_i\|^2 + \|V_i\|$

Highlights

1. Novel Multi-ARIMA Fusion Framework
2. Intelligent Missing Data Compensation mechanism
3. Demonstrated Superior Robustness

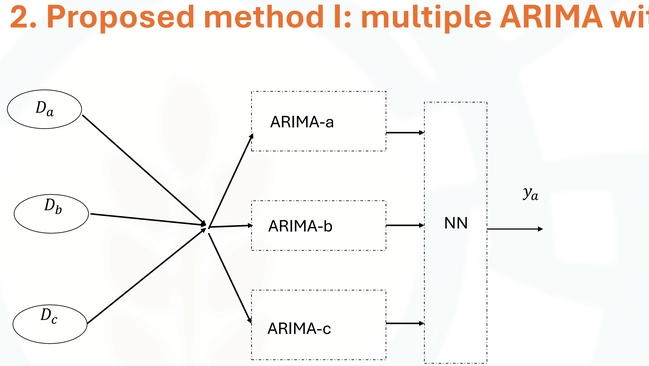
1. Single ARIMA with NN fusion

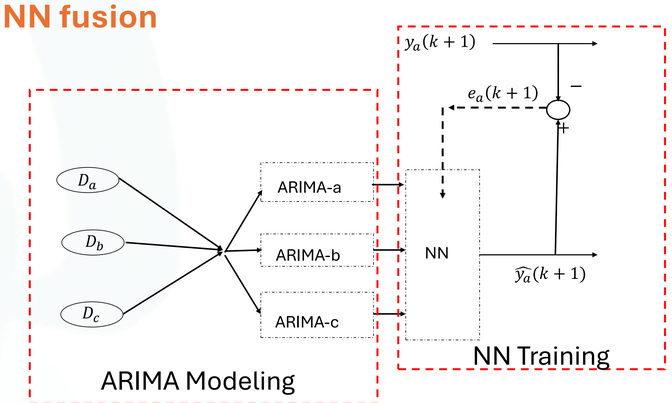




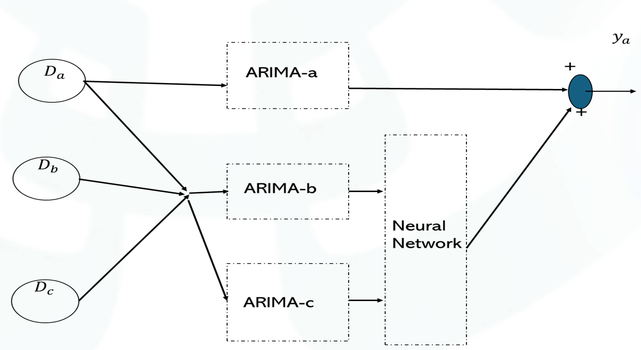
(a) single ARIMA with NN fusion (b) Multiple ARIMA with NN fusion

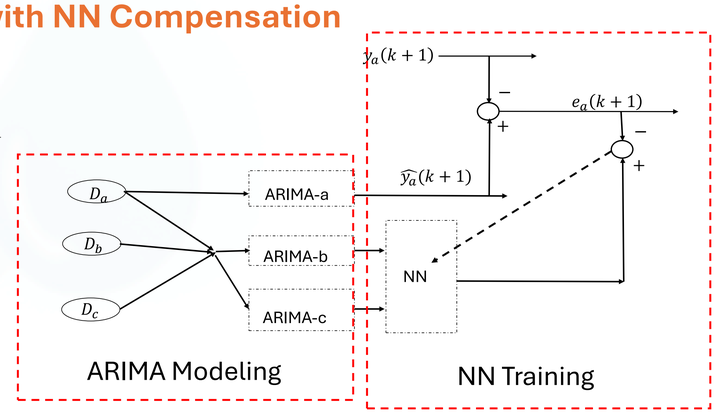
2. Proposed method I: multiple ARIMA with NN fusion





3. Proposed method II: multiple ARIMA with NN Compensation





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